

Serologic Testing for Lyme Disease

Patients with autoimmune diseases (e.g., lupus, rheumatoid arthritis), Epstein-Barr virus infection, bacterial endocarditis, and other tick-borne illnesses are at increased risk for false-positive IgM serologic results. False-positive IgG results may occur in individuals with syphilis, *Helicobacter pylori* infection, systemic lupus erythematosus, ehrlichiosis, and babesiosis.

A newer ELISA test that targets a small peptide from the constant region of the *Borrelia burgdorferi* VlsE protein—known as the C6 peptide test—demonstrates greater specificity compared to traditional whole-cell ELISA. This test detects only IgG antibodies, which develop early in the course of Lyme disease. The sensitivity of the C6 peptide assay in early disease is comparable to that of IgM-based whole-cell ELISA tests.

Western immunoblot assays offer higher specificity by detecting antibodies to individual *B. burgdorferi* antigens. However, there are currently no universally accepted standard serodiagnostic tests or criteria for Lyme disease in Europe or Asia, largely due to the presence of multiple *B. burgdorferi* strains with limited antigen cross-reactivity.

Current Recommendations for Serologic Testing

Due to limitations such as high false-positive rates in the general population, serologic testing should be used selectively and only to support a clinical diagnosis, based on pre-test probability.

- **High pre-test probability:** Patients in endemic areas with classic erythema migrans (EM) lesions should be diagnosed clinically without laboratory testing.
- **Intermediate pre-test probability:** Serologic testing is most useful in patients from endemic regions who present with symptoms suggestive of Lyme disease but lacking diagnostic signs (e.g., facial palsy, arthritis, atrioventricular conduction abnormalities) and without EM.

- **Low pre-test probability:** Testing is not recommended for individuals not in endemic areas, those with non-specific systemic symptoms, or after tick bite without symptoms—false positives outweigh true positives in this group.

Two-Stage Testing Algorithm

1. Begin with a sensitive ELISA (preferably C6 peptide ELISA).
2. If negative and early disease is suspected, repeat testing in 3–4 weeks (early antibiotics may delay antibody response).
3. If ELISA is positive or indeterminate, confirm with Western immunoblot.
4. This two-step approach has 99% to 100% specificity in late Lyme disease.

Interpretation of Results

- In patients with untreated illness lasting >1 month, IgM testing should not be performed.
- A positive IgM blot with a negative IgG blot in this context is considered a negative result due to high false-positive IgM rates.
- Seronegativity in suspected late Lyme disease effectively rules out the diagnosis.
- Persistent IgG or IgM positivity after treatment does not indicate treatment failure, reinfection, or ongoing infection, as antibodies may remain for years.
- No test currently confirms complete clearance of *B. burgdorferi* post-treatment.

Treatment Recommendations

Antibiotic therapy is recommended for all stages of Lyme disease, even though many manifestations resolve spontaneously. Patients may still have symptoms at the end of treatment, but this is not an indication to prolong therapy; symptoms typically improve gradually over time.

Jarisch-Herxheimer reaction—characterized by fever, chills, and worsening arthralgias/myalgias—may occur within hours of the first antibiotic dose and is believed to result from the host's inflammatory response to dying spirochetes.

The Infectious Diseases Society of America (IDSA) has published treatment guidelines for all manifestations of Lyme disease. However, many recommendations are based on expert opinion rather than large, controlled clinical trials.